Value at Risk Analysis

Equibalanced Portfolio: Amazon (AMZN) & Apple (AAPL)

Gabriele Basciu Report 4

1. Portfolio Statistics

The analysis is based on a **6-month window** of daily closing prices for Amazon and Apple, yielding **123 daily return observations**. The portfolio is **equibalanced** with $W_1 = W_2 = \$50$ (total W = \$100). Daily returns are computed as simple percentage changes: $r_t = (P_t - P_{t-1}) / P_{t-1}$.

Statistic	Amazon (AMZN)	Apple (AAPL)
Daily Mean Return	-0.0396%	-0.0120%
Daily Std Dev (σ flat)	1.9625%	1.3310%
Daily Variance	0.000385	0.000177
Correlation (ρ)	0.2451	—
Portfolio σ (equibalanced)	1.3137%	—
Portfolio Mean Return	-0.0258%	—

The correlation of **0.2451** between AMZN and AAPL is positive but moderate, implying meaningful diversification benefits in the joint portfolio. Both assets show negative mean daily returns over the observation window, consistent with a period of market weakness.

2. Parametric VaR — Flat Sigma

Under the **model-building (variance-covariance) approach**, the 1-day VaR for a single asset is: $VaR = W_i \cdot \sigma_i \cdot \sqrt{T} \cdot z_x - \mu_i \cdot T$. *For the joint portfolio, the dollar volatility is:* $\sigma_p(\$) = \sqrt{[(W_1\sigma_1)^2 + (W_2\sigma_2)^2 + 2\rho W_1\sigma_1 W_2\sigma_2]}$.

The flat sigma is estimated as the sample standard deviation over the full 6-month window. Results for T = 1 day:

Level	VaR Amazon ($)	VaR Apple ($)	VaR Sum ($)	VaR Joint ($)	Div. Benefit ($)
95%	1.6140	1.0947	2.7087	2.1609	0.5478
99%	2.2827	1.5482	3.8309	3.0562	0.7747
99.5%	2.5275	1.7143	4.2418	3.3840	0.8578

At the 99% level, the 1-day parametric VaR of the joint portfolio is **\$3.06** — lower than the naïve sum of \$3.83, confirming **diversification benefits of \$0.77**. For T = 10 days, all figures scale by $\sqrt{10} \approx 3.162$ (e.g., joint $VaR_{nn} \approx \$9.66$).

3. Parametric VaR — EWMA (RiskMetrics, λ = 0.94)

RiskMetrics estimates volatility recursively: $\sigma^2_t = \lambda\sigma^2_{t-1} + (1-\lambda)r^2_{t-1}$, with λ = 0.94. Similarly, the covariance is updated as $cov_t = \lambda \cdot cov_{t-1} + (1-\lambda) \cdot r_{1,t-1} \cdot r_{2,t-1}$. The VaR is then computed using the last-day EWMA estimates: $\sigma_1 = 1.9288\%$, $\sigma_2 = 1.2281\%$, $\rho = 0.2766$.

Level	VaR Amazon ($)	VaR Apple ($)	VaR Sum ($)	VaR Joint ($)	Div. Benefit ($)
95%	1.5863	1.0100	2.5963	2.1030	0.4933
99%	2.2435	1.4285	3.6720	2.9743	0.6977
99.5%	2.4841	1.5817	4.0658	3.2933	0.7725

EWMA VaR is slightly **lower** than flat-sigma VaR (e.g., joint 99%: \$2.97 vs \$3.06), suggesting that recent volatility was marginally below the 6-month average. The EWMA approach is **more reactive** to market conditions, making it preferable in rapidly changing environments.

4. Monte Carlo VaR

With N = 10,000 simulations, correlated shocks are drawn as: $r_1 = \sigma_1\sqrt{T}\cdot Z_1$ and $r_2 = \sigma_2\sqrt{T}\cdot(\rho Z_1 + \sqrt{(1-\rho^2)}Z_2)$, where $Z_1, Z_2 \sim N(0,1)$. The portfolio loss per scenario is: Loss = $-(W_1r_1 + W_2r_2)$. VaR is the empirical percentile of the loss distribution.

Level	VaR Joint ($)	N Simulations	Horizon T
95%	2.1416	10,000	1 day
99%	3.0649	10,000	1 day
99.5%	3.3493	10,000	1 day

MC VaR results (e.g., \$3.06 at 99%) are consistent with the parametric flat-sigma approach, as expected when returns are drawn from the same normal distribution used in the parametric model. Minor differences arise from simulation noise.

5 & 6. Historical VaR and Historical Simulation

The **historical VaR (Point 5)** directly uses observed daily losses: $\text{Loss}_t = -(W_1r_{1,t} + W_2r_{2,t})$, and takes the empirical percentile over 123 observations. The **historical simulation (Point 6)** applies past percentage changes to the **current portfolio value**, generating hypothetical scenarios. With a \$100 equibalanced portfolio these approaches yield identical results.

Level	VaR Amazon ($)	VaR Apple ($)	VaR Sum ($)	VaR Joint ($)	Div. Benefit ($)
95%	1.4233	0.9630	2.3863	2.0591	0.3272
99%	2.4346	1.7274	4.1620	3.5591	0.6029
99.5%	2.6059	2.0286	4.6345	3.8407	0.7938

The historical VaR is generally **higher than parametric VaR** at 99% and 99.5% (e.g., joint 99%: \$3.56 vs \$3.06). This reflects **fat tails** in empirical returns — extreme observations in the actual data exceed what a normal distribution would predict, a well-documented stylized fact in equity markets.

7. Additivity Check

Across all methods, the joint VaR is consistently **less than** the sum of individual VaRs — confirming **non-additivity (sub-additivity)**. The diversification benefit arises from the imperfect correlation ($\rho = 0.2451$): when AMZN and AAPL do not move perfectly together, extreme losses are partially offset.

At the 99% flat-sigma level: $VaR_1 + VaR_2 = \$3.83 > VaR_{\wp} = \$3.06$, yielding a **benefit of \$0.77**. If $\rho = 1$, this benefit would vanish. If $\rho < 0$, it would be even larger.

Importantly, sub-additivity holds across all methods *(parametric, EWMA, MC, historical)*. This is notable because VaR is **not guaranteed to be sub-additive** for non-normal distributions.

Conclusions

This report computed VaR for an equibalanced Amazon/Apple portfolio using five methodologies. The **parametric flat-sigma** and **Monte Carlo** methods produce consistent results, grounded in the normality assumption. The **EWMA approach** captures time-varying volatility, proving more reactive to recent market conditions. **Historical methods** reveal heavier tails than the normal model assumes, producing higher tail-risk estimates at elevated confidence levels. In all cases, the joint VaR is sub-additive, confirming the diversification benefit of combining two imperfectly correlated assets.